

Real-Time Financial Fraud Detection with Learned Hybrid Gating and Uncertainty-Aware Triage

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Abstract—Financial fraud detection is challenging because fraudulent transactions are rare, behavior evolves over time, and operational systems must distinguish confident predictions from cases requiring human investigation. We present an end-to-end fraud detection and uncertainty-triage system that combines supervised and unsupervised deep learning. An FT-CAT Transformer produces supervised fraud probabilities from current and historical transaction information, while a denoising autoencoder provides an independent reconstruction-based anomaly signal. A learned gating network dynamically combines these signals with transaction-history activity and amount-intensity features. Split conformal prediction converts the fused fraud score into operational prediction sets for automatic approval, automatic blocking, or human review. On the chronological held-out test set, the FT-CAT model achieved a PR-AUC of 0.3259 and ROC-AUC of 0.8172. At a target miscoverage level of 0.01, the conformal system achieved 98.99% empirical marginal coverage with a 95% bootstrap confidence interval of 98.92–99.06%, while assigning 23.19% of transactions to human review. CPU benchmarking measured a median batch-1 neural inference latency of 6.56 ms and peak tested throughput of approximately 2,746 transactions per second at batch size 128. The system additionally provides SHAP-based anomaly explanations, a Streamlit analyst dashboard, FastAPI serving, and EXIR and ONNX model serialization. Repository: <https://github.com/sa6uhi/DL-final-project>.

Index Terms—financial fraud detection, deep learning, transformer, anomaly detection, conformal prediction, uncertainty quantification

I. Introduction

Financial transaction systems must identify fraudulent behavior under severe class imbalance while operating under strict latency and reliability requirements. A useful fraud-detection product must therefore do more than produce a binary prediction. It should combine complementary evidence, expose uncertain cases to analysts, provide interpretable information about suspicious transactions, and remain deployable on modest hardware. Purely supervised fraud classifiers can learn discriminative patterns from historical labels but may struggle when fraudulent behavior differs from previously observed examples. Conversely, unsupervised anomaly detectors can identify unusual transaction structure without relying exclusively on fraud labels, but unusual behavior is not necessarily fraudulent. This motivates a hybrid architecture in which supervised and unsupervised signals contribute jointly to the final decision. We develop an end-to-end fraud triage engine with four main components. First, an FT-CAT Transformer models the supervised probability of fraud using

current transaction attributes and recent transaction history. Second, a denoising autoencoder trained on legitimate behavior produces a reconstruction-based anomaly score. Third, a learned gating network dynamically fuses the supervised probability, normalized anomaly score, history activity, and historical amount intensity. Finally, split conformal prediction transforms the fused score into prediction sets that support automatic approval, automatic blocking, and human review. The system is designed as an Industry Product prototype rather than only an offline classifier. It includes a FastAPI inference service, a Streamlit analyst dashboard, SHAP-based explanations, model serialization, automated tests, and CPU inference benchmarking. The main contributions of this work are:

- an end-to-end hybrid fraud-detection architecture combining supervised FT-CAT prediction with unsupervised DAE anomaly detection;
- a learned gating network that dynamically combines model signals and causal transaction-history context instead of relying only on a fixed fusion coefficient;
- uncertainty-aware operational triage using split conformal prediction;
- explainability through SHAP analysis of the DAE anomaly component;
- deployment-oriented evaluation including CPU latency, throughput, model serialization, API serving, and an analyst-facing dashboard.

The remainder of this paper describes related work, the proposed architecture, the leakage-aware experimental protocol, empirical results, deployment characteristics, limitations, and conclusions.

II. Related Work

Fraud detection is commonly formulated as a highly imbalanced binary classification problem in which the minority fraud class is operationally more important than overall accuracy. Tree-based methods and neural classifiers are frequently used to model nonlinear interactions among transaction attributes, while precision-recall metrics are particularly informative under strong class imbalance. Transformer architectures have increasingly been adapted to structured and tabular data [2]. Attention mechanisms [1] allow models to learn interactions among features and, when transaction histories are available, relationships between a current transaction and previous activity. Our supervised component follows this

general direction by combining tabular transaction information with historical context. Autoencoders provide a complementary approach to fraud detection. When trained primarily or exclusively on legitimate examples, reconstruction error can act as an anomaly score because transactions inconsistent with learned legitimate structure may reconstruct poorly. However, anomaly is not equivalent to fraud. Our system therefore treats reconstruction error as one signal rather than as the final decision. Conformal prediction provides a model-agnostic framework for constructing prediction sets with finite-sample marginal coverage under exchangeability [4], [5]. Instead of forcing every transaction into a single predicted class, prediction sets can distinguish confident cases from ambiguous cases. We use this property to map uncertainty directly to an operational human-review queue.

III. Method

A. Problem Formulation

For each transaction x_i , the objective is to estimate whether its binary label $y_i \in \{0, 1\}$ corresponds to a legitimate or fraudulent transaction. The system produces a fused fraud probability

$$p_i = f_\theta(x_i, h_i), \quad (1)$$

where h_i represents causal historical information available before the current transaction. Instead of using p_i only for hard binary classification, the final stage constructs a conformal prediction set

$$\Gamma_\alpha(x_i) \subseteq \{0, 1\}, \quad (2)$$

where α is the target miscoverage level.

B. FT-CAT Transformer

The supervised branch uses an FT-CAT Transformer to estimate fraud probability. The model consumes current transaction information together with a fixed recent history window. Continuous and categorical information is represented in a learned feature space, while attention mechanisms [1] model interactions between the current transaction and its historical context. The supervised output is

$$P_{\text{FT}} = P(y = 1 \mid x, h). \quad (3)$$

This probability forms one of the primary inputs to the learned fusion stage.

C. Denoising Autoencoder

The unsupervised branch is a denoising autoencoder trained on legitimate transactions. Its encoder compresses an 800-dimensional numeric feature vector through

$$800 \rightarrow 256 \rightarrow 128 \rightarrow 32, \quad (4)$$

followed by a symmetric decoder

$$32 \rightarrow 128 \rightarrow 256 \rightarrow 800. \quad (5)$$

The network uses batch normalization and LeakyReLU activations. During training, feature corruption encourages the autoencoder to learn robust structure rather than simply copy

the input. For transaction x , the anomaly signal combines the squared reconstruction residual with an L_1 term that discourages a small number of large, concentrated residuals from dominating the score,

$$S_{\text{anomaly}}(x) = \sum_{j=1}^d (x_j - \hat{x}_j)^2 + \gamma_1 \sum_{j=1}^d |x_j - \hat{x}_j|. \quad (6)$$

The parameter $\gamma_1 = 0.4$ is fixed by configuration. The anomaly score is normalized by its 99.9th training-set percentile before entering the gating network.

D. Learned Hybrid Gating

A fixed weighted combination of anomaly and supervised scores assumes that the same mixture is appropriate for every transaction. We instead learn the fusion function. The gating network receives four inputs,

$$z = [\tilde{S}_{\text{anomaly}}, P_{\text{FT}}, D_{\text{hist}}, A_{\text{hist}}], \quad (7)$$

where $\tilde{S}_{\text{anomaly}}$ is the normalized DAE anomaly score, D_{hist} is history density, and A_{hist} represents historical amount intensity. History density is the fraction of active entries in the recent transaction window. Historical amount intensity is based on the logarithmically transformed mean absolute transaction amount among active historical rows. These quantities represent causal transaction-history activity and intensity rather than future-looking features. The learned gating network maps these inputs to the final fraud probability,

$$P_{\text{gate}} = g_\phi(z). \quad (8)$$

E. Split Conformal Triage

The fused gate probability is passed to a split conformal predictor [4], [5]. Calibration examples are kept separate from the examples used to fit the upstream models and gating network. For class y , a probability-based nonconformity score can be written as

$$s(x, y) = 1 - \hat{P}(y \mid x). \quad (9)$$

The calibration scores determine a finite-sample corrected quantile. Labels whose nonconformity scores fall below the calibrated threshold are included in the prediction set. The resulting operational policy is

$$\Gamma_\alpha(x) = \begin{cases} \{0\}, & \text{Auto-Approve,} \\ \{1\}, & \text{Auto-Block,} \\ \{0, 1\}, & \text{Human Review,} \\ \emptyset, & \text{Human Review.} \end{cases} \quad (10)$$

Mapping the empty set to review is a defensive product decision: a transaction for which neither class is supported should not be silently approved or blocked.

F. Explainability

SHAP [3] is used to analyze the DAE anomaly component. A local waterfall visualization explains feature contributions for a selected high-anomaly transaction, while global analysis summarizes feature importance across sampled transactions. These explanations apply specifically to the DAE reconstruction-anomaly component. They should not be interpreted as explanations of the complete learned-gate decision.

IV. Experimental Setup

A. Temporal Evaluation Protocol

Transactions are ordered chronologically to reduce temporal leakage. The original dataset is divided into approximately 70% training, 15% validation, and 15% final out-of-time testing blocks, while equal timestamps at split boundaries are kept together. Additional subdivision is required because the learned gate, upstream model early stopping, and conformal calibration cannot safely reuse the same labels. The original training block is therefore subdivided into upstream model training and early-stopping subsets. The original validation period is reserved for gate development and conformal calibration. Gate development is itself separated into gate-training and gate early-stopping subsets. This produces approximately 63% upstream-model training, 7% upstream-model early stopping, 6% gate training, 1.5% gate early stopping, 7.5% conformal calibration, and 15% final testing. Exact proportions can vary slightly because identical boundary timestamps remain grouped. The final chronological test set contains 88,581 transactions and is not used for model fitting or conformal calibration.

B. Evaluation Metrics

Because fraud is highly imbalanced, accuracy alone is insufficient. Supervised and gating models are evaluated primarily using area under the precision-recall curve (PR-AUC), ROC-AUC, and true-positive rate at the configured false-positive-rate operating point. Conformal evaluation reports empirical marginal coverage, class-conditional coverage, review rate, singleton rate, review precision, reviewed-fraud capture, Brier score, and expected calibration error (ECE). Deployment evaluation reports latency percentiles and throughput over several batch sizes.

C. Implementation and Reproducibility

The system is implemented in Python and PyTorch. Model and experiment configuration is centralized in configuration files, and fixed random seeds are used for reproducibility. Tests cover the gating network, conformal predictor, SHAP integration, dashboard behavior, and serialization. The trained DAE, FT-CAT, and learned gate can be exported to PyTorch EXIR and ONNX. Export routines validate numerical parity against PyTorch inference.

TABLE I
FT-CAT performance on the held-out chronological test set.

Metric	Test Result
PR-AUC	0.3259
ROC-AUC	0.8172
TPR @ configured FPR	0.2488
Loss	0.0119

TABLE II
Learned gate robustness across random seeds.

Seed	PR-AUC	ROC-AUC	TPR
42	0.3420	0.8373	0.2293
123	0.3293	0.8312	0.2348
2026	0.3432	0.8324	0.2431
Mean	0.3381	0.8336	0.2357
SD	0.0077	0.0032	0.0070

V. Results

A. FT-CAT Performance

Table I reports final FT-CAT performance on the chronological held-out test set. The lower test PR-AUC relative to development validation indicates that out-of-time generalization is materially harder than performance measured within the development period. This reinforces the importance of chronological evaluation for fraud detection.

B. Learned Gate Robustness

The learned gating model was evaluated across three fixed random seeds on the development-validation data. Results are shown in Table II. The relatively small standard deviations indicate that the learned fusion model is not strongly dependent on a single random initialization. Seed 42 was used for the production checkpoint according to the project's predetermined random seed rather than selected post hoc from the three results.

C. Conformal Triage

At $\alpha = 0.01$, split conformal prediction targets 99% finite-sample marginal coverage under exchangeability. Table III reports performance on the chronological held-out test set. The empirical marginal coverage of 98.99% is close to the nominal target. However, aggregate coverage masks substantial class asymmetry: legitimate coverage is 100%, while fraud coverage is approximately 70.94%. The review queue captures approximately 70.94% of fraudulent test transactions. No transactions were assigned to the auto-block category at this conservative operating point. Under the chosen 99% coverage target, the system therefore behaves primarily as an automatic-approval plus human-review mechanism. Figure 1 illustrates the operational trade-off between predictive coverage and analyst workload across the evaluated conformal operating points. More conservative coverage requirements increase the fraction of transactions requiring human review, while less conservative settings can reduce review workload at the cost of weaker

TABLE III
Conformal triage results at $\alpha = 0.01$.

Metric	Result
Target coverage	99.00%
Empirical coverage	98.99%
95% bootstrap CI	98.92–99.06%
Legitimate-class coverage	100.00%
Fraud-class coverage	70.94%
Human-review rate	23.19%
Auto-approve rate	76.81%
Auto-block rate	0.00%
Review precision	10.65%
Reviewed-fraud capture	70.94%
Brier score	0.0319
ECE	0.0121

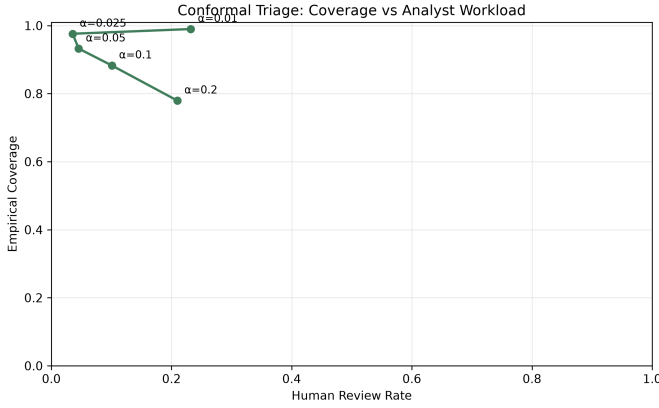


Fig. 1. Empirical conformal coverage and human-review workload across evaluated miscoverage levels.

coverage. At the selected $\alpha = 0.01$ operating point, the system achieves 98.99% empirical marginal coverage while routing 23.19% of test transactions to review. The conformal layer therefore acts not only as an uncertainty estimator but also as an operational mechanism for controlling analyst workload.

D. Autoencoder Latent-Space Analysis

A t-SNE analysis was performed on 5,000 held-out transactions using the 32-dimensional DAE bottleneck representation. The resulting class-based silhouette score was 0.1095. The visualization in Fig. 2 shows partial structural separation but substantial overlap between legitimate and fraudulent transactions. The weak but non-zero separation is consistent with the role of the DAE as an unsupervised anomaly detector rather than a standalone fraud classifier. The representation therefore provides complementary structural information rather than a clean fraud decision boundary.

E. SHAP Analysis

SHAP analysis was used to inspect the DAE anomaly component. Across independent global samples, the mean Spearman correlation of feature rankings was 0.8924, while mean top-feature Jaccard similarity was 0.4762. Figure 3 shows the global SHAP importance profile for the DAE anomaly

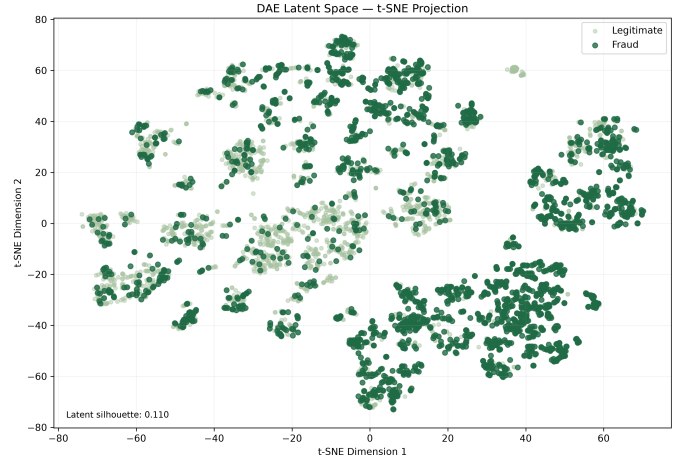


Fig. 2. t-SNE projection of the 32-dimensional DAE bottleneck representation.

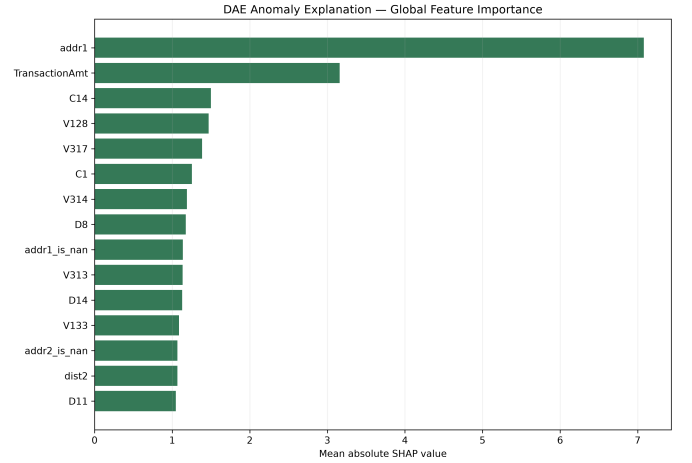


Fig. 3. Global SHAP feature importance for the DAE reconstruction-anomaly component.

component. In the sampled analysis, `addr1` receives the strongest global anomaly attribution. The high mean Spearman rank correlation indicates that the broader ordering of feature importance is relatively stable across independent samples, although the lower mean top-feature Jaccard similarity shows that membership of the exact top-feature set is less stable. Importantly, these attributions explain reconstruction anomaly rather than the final fused fraud probability. The prominence of `addr1` therefore should not be interpreted as evidence that it is the strongest predictor of fraud.

F. CPU Inference Performance

The neural inference path was benchmarked on CPU using 10 warm-up iterations and 50 measured iterations per batch size. The benchmark includes the DAE, FT-CAT, transaction-history feature assembly, and learned gate. It excludes HTTP/Pydantic overhead, disk I/O, SHAP computation, and conformal decision logic. Batch size 128 produced the highest tested throughput at approximately 2,746 transactions

TABLE IV
End-to-end neural CPU inference benchmark.

Batch	Median (ms)	P99 (ms)	Throughput
1	6.56	26.95	128 tx/s
32	14.90	24.08	2,092 tx/s
128	44.94	67.56	2,746 tx/s
512	189.73	290.84	2,693 tx/s

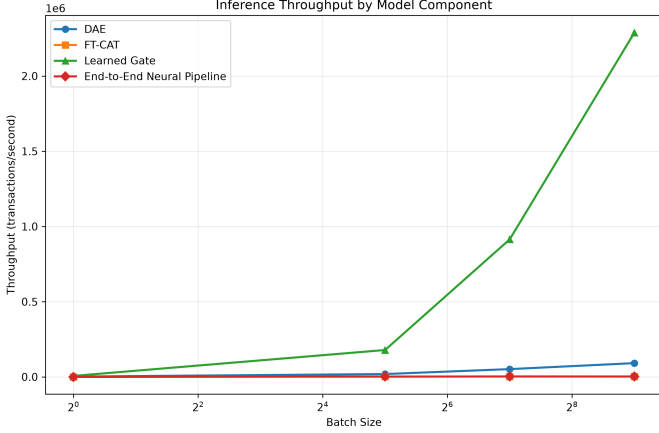


Fig. 4. Measured neural inference throughput across evaluated batch sizes on CPU.

per second. At batch size 1, median neural inference latency was 6.56 ms. Component-level profiling showed that FT-CAT dominates the neural compute path, whereas learned-gate inference itself is lightweight. Figure 4 shows that throughput increases substantially with batching before reaching its highest tested value at batch size 128. Throughput rises from approximately 128 transactions per second at batch size 1 to 2,746 transactions per second at batch size 128, before remaining approximately flat at 2,693 transactions per second for batch size 512. Moderate batching therefore provides substantial computational efficiency, while increasing the batch size beyond 128 produced no additional throughput benefit in the tested CPU environment. These measurements describe model-side neural inference and should not be interpreted as complete HTTP API latency.

G. Deployment and Serialization

The trained DAE, FT-CAT, and learned gate were successfully serialized to both PyTorch EXIR and ONNX representations. Export-time parity checks compare serialized outputs with their PyTorch reference implementations. The product additionally exposes a FastAPI scoring interface and a Streamlit analyst dashboard. The dashboard surfaces prediction, batch analysis, model insights, conformal triage, explainability, and artifact-readiness information.

VI. Discussion and Limitations

The experiments demonstrate the value of treating fraud detection as a triage problem rather than forcing every transaction into an automatic binary decision. The supervised FT-CAT

branch provides discriminative fraud evidence, while the DAE contributes a complementary anomaly signal. Learned gating provides a flexible mechanism for combining these signals with transaction history. The conformal results expose an important operational trade-off. The system achieves 98.99% empirical marginal coverage on the chronological test set at the 99% target, but approximately 23.19% of transactions require human review. At this conservative operating point, no test transactions are automatically blocked. Therefore, the current configuration is better characterized as a high-coverage approval-and-review system than as a fully automated fraud-blocking system. A second limitation is class-conditional behavior. Marginal coverage is close to the nominal target, but fraud-class coverage is substantially lower than legitimate-class coverage. Marginal split conformal prediction does not guarantee equal class-conditional coverage. A third limitation concerns the assumptions behind conformal prediction. Finite-sample marginal coverage applies under exchangeability between calibration and future examples. Financial transaction data are chronological and may undergo temporal or adversarial distribution shift. Consequently, 98.99% should be reported as observed chronological test coverage, not as an unconditional guarantee for future production traffic. The SHAP implementation is also intentionally scoped. Current explanations describe the DAE anomaly component and do not provide a complete attribution of the final learned-gate decision. Extending explainability to the complete fused system is an important direction for future work. The fixed-coefficient and learned-gating comparison also requires careful interpretation. The learned gate incorporates transaction-history activity and amount-intensity features in addition to the DAE and FT-CAT signals, whereas a simple fixed-weight fusion uses only the two model scores. Improvements from the learned system therefore cannot be attributed exclusively to nonlinear fusion; the additional causal history context is also part of the learned-gate design. Finally, the latency benchmark measures the neural computation path rather than complete request-to-response service latency. Production deployment would require additional benchmarking of network overhead, request validation, concurrency, monitoring, and data retrieval.

VII. Conclusion

We presented an end-to-end financial fraud detection and uncertainty-triage engine combining an FT-CAT Transformer, a denoising autoencoder, learned hybrid gating, and split conformal prediction. The system achieved 98.99% empirical marginal coverage at a nominal 99% target on the chronological held-out test set while routing 23.19% of transactions to human review. The results demonstrate the practical role of uncertainty-aware triage: rather than forcing every prediction into an automated action, the system explicitly identifies cases requiring analyst intervention. The deployment prototype supports FastAPI serving, an analyst-facing Streamlit dashboard, SHAP-based anomaly explanations, and EXIR and ONNX serialization. CPU benchmarking demonstrated a median batch-1 neural inference latency of 6.56 ms and

peak tested throughput of approximately 2,746 transactions per second. Future work should investigate class-conditional conformal calibration, distribution-shift monitoring, fused-model explanations, and alternative operating points that support safe automatic blocking.

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